

NOTE:

- These are not exact final questions. These questions are intended to review the material.

1 Quick Questions

- (a) Consider a set A of positive integers and an integer $a \in A$. The set A contains a subset that adds up to t if and only if the set _____ contains a subset that adds up to _____ or _____.
- (b) **T or F:** For every dynamic program, we can assign weights to edges in the directed acyclic graph of dependencies among subproblems, such that finding a shortest path in this DAG is equivalent to solving the dynamic program.
- (c) In a BFS traversal of a graph, if a node v is discovered at layer k , the earliest a node can be revisited (if at all) is at layer _____.
- (d) A loop invariant property of is that DFS at the discovery of any node v , the nodes in the stack represent the path from the _____ to _____.
- (e) **T or F:** Given a directed graph $G = (V, E)$, run BFS from a vertex $s \in V$. While processing a vertex u , if some $v \in \text{Adj}(u)$ has already been processed, then G contains a directed cycle.

2 Searching Deep

Assume that we have run Depth-First Search (DFS) on a directed graph $G(V, E)$ and obtained the following discovery and finishing times for each vertex in V :

Vertex	v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8	v_9	v_{10}
d	1	3	4	5	7	10	11	12	17	18
f	2	16	9	6	8	15	14	13	20	19

Answer the following questions assuming that we do not know E . Note that these questions are independent of each other and a fact given in a question does not necessarily apply to other questions. **Briefly justify your answers.**

- (a) (8 points) Can this graph be strongly connected?
- (b) (8 points) List all outgoing edges of v_4 that could exist and classify those edges (tree, forward, back, cross) with respect to this run of DFS assuming that they exist.
- (c) (8 points) Given that $v_5.d = 1$ in some run of DFS, what are the minimum and maximum possible values of $v_5.f$ in that run of DFS?
- (d) (8 points) True or False: If another run of DFS returns $v_7.f > v_6.f$, then there must be a cycle that contains v_6 and v_7 .

3 Presentation Question: Graph B

NOTE: For graph algorithms you need to provide following items:

- Representation: How are you constructing the graph?
- Pseudocode
- Proof of correctness
- Run time analysis

Problem Statement: We have n gifts, and each gift has a value. Gift i has value v_i . We need to divide the gifts between people. Every gift has one person that wants it. But, every person was given a random gift. Given a list of pairs (i, j) meaning person i can pass a gift on to person j , see if (*regardless of where the gifts were initially distributed*) each gift can be passed to the person who wants it. Devise an efficient algorithm to do so. You do not need to return the actual sequence of exchanges.

Example

Input:

$n = 4$ (4 people, 4 gifts)

List of pairs (i, j) (person i can pass to person j):

$(1, 2), (2, 3), (3, 4), (4, 1)$

Output:

True

4 Presentation Question: Greedy Algorithm

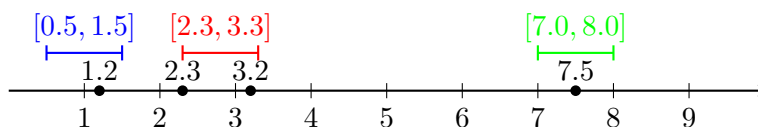
NOTE: For greedy algorithm problems you need to provide following items:

- Greedy Choice
- Proof of Greedy Choice
- Pseudocode
- Run time analysis

Problem Statement: Given a set of real-valued points, $P = \{x_1, \dots, x_n\}$, describe an efficient algorithm that determines the smallest set of unit-length closed intervals, $[a, a + 1]$, that contain all of the given points.

Example

Input: $P = [1.2, 2.3, 3.2, 7.5]$



Output:

of intervals = 3

$I = \{[0.5, 1.5], [2.3, 3.3], [7, 8]\}$

Important Note: the example above is an optimal solution to the problem, but does not necessarily follow the greedy choice. You will have to figure out the correct greedy choice!